

Shrihari Nagesh_PES1UG24CS445-BPS#17

Kanban Project — Backlog Submission (Kanban_BPS#17)

Problem Statement #17 — Healthcare & Telemedicine: *Vaccination Cohort & Dose Scheduling System*

A public health administration platform to organize demographic vaccination rollouts, enforce minimum interval days between doses, schedule center slots, and generate verifiable QR digital vaccination certificates.

Actors: Citizen Registrant, Vaccination Officer

1. Requirements (5 FR + 2 NFR)

The table below is the full requirements set from Lab 1, reused as the raw material for the Kanban board's Epics and Stories.

ID	Priority / Type	Description	Acceptance Criteria
FR-001	Priority: High	The system shall enforce dose interval rules (e.g., minimum 28 days after Dose 1) before unlocking Dose 2 booking for a citizen.	Pass: Dose 2 booking blocked if interval < 28 days. Fail: Early vaccination slot confirmed.
FR-002	Priority: High	The system shall allow a Citizen Registrant to create a profile with demographic details (age, comorbidity flags, occupation category) and automatically assign them to an eligible vaccination cohort/phase.	Pass: Registrant is auto-tagged with the correct cohort based on age/comorbidity rules. Fail: A registrant is placed in an ineligible cohort.
FR-003	Priority: High	The system shall let a Citizen Registrant search nearby vaccination centers by pincode/date and book an available time slot within that center's daily capacity.	Pass: Booking succeeds only while booked count < center capacity, and the slot count decrements. Fail: Center accepts a booking beyond its capacity.
FR-004	Priority: Medium	The system shall allow a Vaccination Officer to check in a citizen, record the administered dose (vaccine brand, batch/lot number, date), and update the citizen's vaccination status.	Pass: Citizen status moves from "Slot Booked" to "Dose Administered" with a batch number stored. Fail: Status can be updated without a batch number.
FR-005	Priority: Medium	Upon completion of the required dose count, the system shall auto-generate a digitally signed QR vaccination certificate for the citizen, downloadable as a PDF.	Pass: Certificate QR encodes citizen ID, dose dates, and a digital signature. Fail: A certificate is generated before the final dose is administered.
NFR-001	Type: Performance & Security Priority: High	The vaccination certificate verification endpoint shall authenticate digitally signed QR codes offline/online in under 150 ms.	Pass: Benchmarking tests confirm target latency and security standards under simulated peak load.
NFR-	Type:	All citizen personal and health data	Pass: A security review

ID	Priority / Type	Description	Acceptance Criteria
002	Security & Privacy Priority: High	(name, ID, vaccination history) shall be encrypted at rest and in transit (TLS 1.2+/AES-256) and accessible only to authorized Vaccination Officers.	confirms encryption at rest/in transit and role-based access control. Fail: PII is retrievable without authentication or found in plaintext.

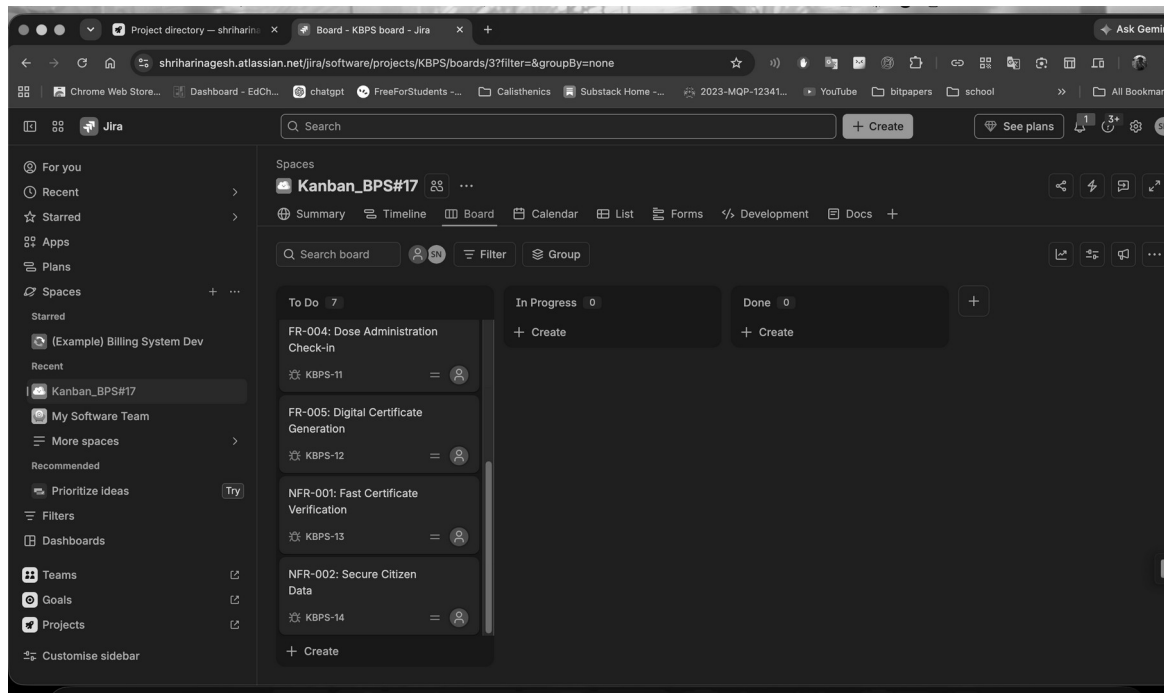


Figure 1. Kanban_BPS#17 board (KBPS project) with all 7 requirement work items (FR-001...FR-005, NFR-001, NFR-002) added to the To Do column.

2. Epics

Lab 1's functional themes were grouped into three Epics:

- Epic 1: Citizen Registration & Eligibility — Enable citizens to register their demographic details and be automatically placed into the correct vaccination cohort/phase.
- Epic 2: Vaccination Scheduling & Administration — Cover dose-interval enforcement, center slot booking, and officer-side dose administration and record-keeping.
- Epic 3: Certificate Generation & Data Security — Cover the digital certificate issuance workflow and the performance/security guarantees around citizen data.

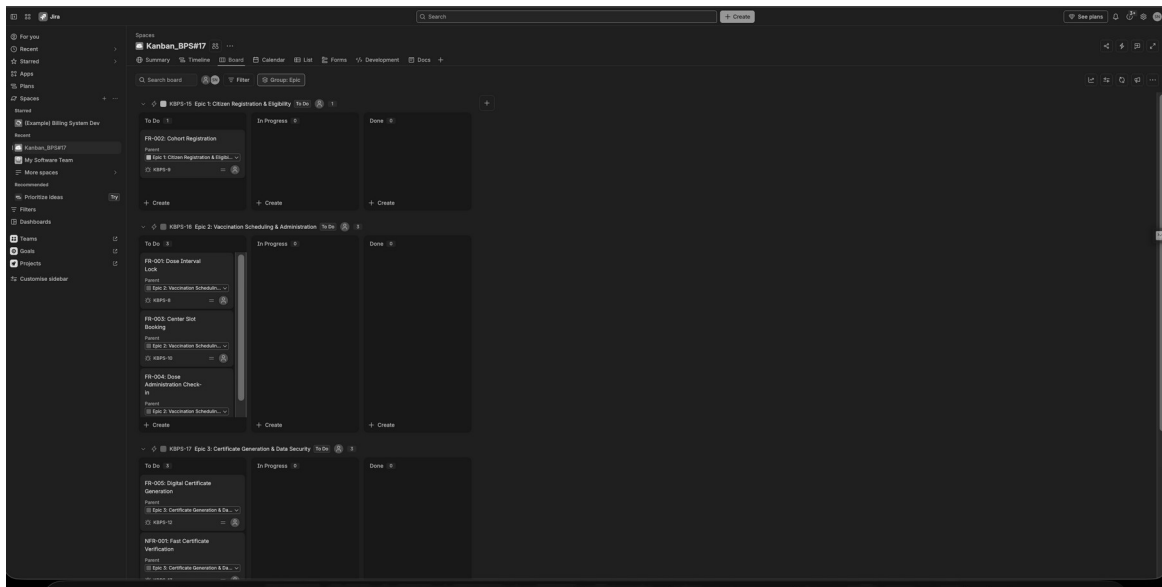


Figure 2. Board grouped by Epic (KBPS-15, KBPS-16, KBPS-17), each requirement's work item linked to its Epic via the Parent field: Epic 1 ← FR-002; Epic 2 ← FR-001, FR-003, FR-004; Epic 3 ← FR-005, NFR-001, NFR-002.

3. User Stories for all 7 Requirements

Each of the 7 requirements (FR-001...FR-005, NFR-001, NFR-002) becomes one User Story, written in As a / I want / So that form, with supporting Tasks and Sub-tasks.

Epic 1: Citizen Registration & Eligibility

Description: Enable citizens to register their demographic details and be automatically placed into the correct vaccination cohort/phase.

Story 1.1: Cohort Registration (FR-002)

As a Citizen Registrant, I want to register my demographic details and get assigned to my eligible vaccination cohort, so that I only see vaccination options I actually qualify for.

Priority: High | Story Points: 5

- Task: Build the citizen registration form UI (name, age, comorbidity flags, occupation).
- Task: Implement the cohort-eligibility rule engine.
- Sub-task: Validate age/comorbidity inputs on submit.
- Sub-task: Unit test cohort-assignment logic.

Epic 2: Vaccination Scheduling & Administration

Description: Cover dose-interval enforcement, center slot booking, and officer-side dose administration and record-keeping.

Story 2.1: Dose Interval Lock (FR-001)

As a Citizen Registrant, I want the system to block me from booking Dose 2 too early, so that I follow the medically required interval between doses.

Priority: Highest | Story Points: 3

- Task: Implement the dose-interval calculation service.
- Task: Add a booking validation check that reads the interval rule.
- Sub-task: Write unit tests for interval edge cases (27 vs 28 days).
- Sub-task: Add a UI warning message when a booking is locked.

Story 2.2: Center Slot Booking (FR-003)

As a Citizen Registrant, I want to search nearby centers and book an available slot, so that I can get vaccinated at a convenient time and place.

Priority: High | Story Points: 8

- Task: Build the center search API (filter by pincode/date).
- Task: Build the slot-capacity and booking engine.
- Sub-task: Implement capacity decrement as an atomic transaction.
- Sub-task: Add a booking confirmation notification.

Story 2.3: Dose Administration Check-in (FR-004)

As a Vaccination Officer, I want to check in a citizen and record their administered dose details, so that the citizen's vaccination record stays accurate and up to date.

Priority: Medium | Story Points: 5

- Task: Build the officer check-in dashboard.
- Task: Implement the dose/batch-number recording form.
- Sub-task: Add batch-number validation (reject empty values).
- Sub-task: Update the citizen status state machine.

Epic 3: Certificate Generation & Data Security

Description: Cover the digital certificate issuance workflow and the performance/security guarantees around citizen data.

Story 3.1: Digital Certificate Generation (FR-005)

As a Citizen Registrant, I want to receive a digitally signed QR certificate after my final dose, so that I have verifiable proof of vaccination.

Priority: Medium | Story Points: 8

- Task: Build the certificate-generation service.
- Task: Integrate digital signing of the certificate payload.
- Sub-task: Generate the QR encoding (citizen ID + dose dates + signature).
- Sub-task: Implement PDF export.

Story 3.2: Fast Certificate Verification (NFR-001)

As a Vaccination Officer, I want the certificate verification scan to return a result in under 150 ms, so that I can process citizens quickly at checkpoints without delays.

Priority: High | Story Points: 5

- Task: Optimize the QR signature-verification service.
- Task: Load-test the verification endpoint under peak conditions.
- Sub-task: Benchmark response time under simulated peak load.
- Sub-task: Add caching for public keys used in signature checks.

Story 3.3: Secure Citizen Data (NFR-002)

As a Vaccination Officer, I want citizen personal and health data to be encrypted and access-controlled, so that sensitive medical information stays protected from unauthorized access.

Priority: Highest | Story Points: 5

- Task: Implement AES-256 encryption at rest for citizen records.
- Task: Enforce TLS 1.2+ for all data in transit.

- Sub-task: Set up role-based access control for officer accounts.
- Sub-task: Run a security-audit checklist against the endpoints.

Work	Assignee	Reporter	Priority	Status	Resolution	Created	Updated	Due date
KBPS-15 Epic 1: Citizen Registration & Eligibility	Unassigned	Shrihari Nagesh	Medium	To Do	Unresolved	30 Aug 2026, 15:19	30 Aug 2026, 15:19	None
KBPS-19 FR-002: Cohort Registration	Unassigned	Shrihari Nagesh	Medium	To Do	Unresolved	30 Aug 2026, 15:15	30 Aug 2026, 15:33	None
KBPS-19 Story 1.1: Cohort Registration	Unassigned	Shrihari Nagesh	Medium	To Do	Unresolved	30 Aug 2026, 15:39	31 Aug 2026, 09:02	None
KBPS-16 Epic 2: Vaccination Scheduling & Administration	Unassigned	Shrihari Nagesh	Medium	To Do	Unresolved	30 Aug 2026, 15:19	30 Aug 2026, 15:19	None
KBPS-18 FR-001: Dose Interval Lock	Unassigned	Shrihari Nagesh	Medium	To Do	Unresolved	30 Aug 2026, 15:15	30 Aug 2026, 15:33	None
KBPS-10 FR-003: Center Slot Booking	Unassigned	Shrihari Nagesh	Medium	To Do	Unresolved	30 Aug 2026, 15:15	30 Aug 2026, 15:33	None
KBPS-11 FR-004: Dose Administration Check-in	Unassigned	Shrihari Nagesh	Medium	To Do	Unresolved	30 Aug 2026, 15:16	30 Aug 2026, 15:34	None
KBPS-20 Story 2.1: Dose Interval Lock	Unassigned	Shrihari Nagesh	Medium	To Do	Unresolved	30 Aug 2026, 15:41	31 Aug 2026, 09:02	None
KBPS-21 Story 2.3: Dose Administration Check-in	Unassigned	Shrihari Nagesh	Medium	To Do	Unresolved	30 Aug 2026, 15:44	31 Aug 2026, 09:03	None
KBPS-25 Story 2.2: Center Slot Booking	Unassigned	Shrihari Nagesh	Medium	To Do	Unresolved	30 Aug 2026, 15:47	31 Aug 2026, 09:09	None
KBPS-17 Epic 3: Certificate Generation & Data Security	Unassigned	Shrihari Nagesh	Medium	To Do	Unresolved	30 Aug 2026, 15:20	30 Aug 2026, 15:20	None
KBPS-12 FR-005: Digital Certificate Generation	Unassigned	Shrihari Nagesh	Medium	To Do	Unresolved	30 Aug 2026, 15:16	30 Aug 2026, 15:34	None
KBPS-13 NFR-001: Fast Certificate Verification	Unassigned	Shrihari Nagesh	Medium	To Do	Unresolved	30 Aug 2026, 15:16	30 Aug 2026, 15:34	None
KBPS-14 NFR-002: Secure Citizen Data	Unassigned	Shrihari Nagesh	Medium	To Do	Unresolved	30 Aug 2026, 15:16	30 Aug 2026, 15:34	None
KBPS-22 Story 3.1: Digital Certificate Generation	Unassigned	Shrihari Nagesh	Medium	To Do	Unresolved	30 Aug 2026, 15:44	31 Aug 2026, 09:05	None
KBPS-23 Story 3.2: Fast Certificate Verification	Unassigned	Shrihari Nagesh	Medium	To Do	Unresolved	30 Aug 2026, 15:44	31 Aug 2026, 09:09	None
KBPS-24 Story 3.3: Secure Citizen Data	Unassigned	Shrihari Nagesh	Medium	To Do	Unresolved	30 Aug 2026, 15:44	31 Aug 2026, 09:09	None

Figure 3. List view showing all 7 User Stories (KBPS-19 to KBPS-25) created and linked via Parent to their Epics, alongside each Epic's FR/NFR requirement — Epic 1: FR-002 + Story 1.1; Epic 2: FR-001/FR-003/FR-004 + Story 2.1/2.2/2.3; Epic 3: FR-005/NFR-001/NFR-002 + Story 3.1/3.2/3.3.

Work	Assignee	Reporter	Priority	Status	Resolution
KBPS-15 Epic 1: Citizen Registration & Eligibility	Unassigned	Shrihari Nagesh	Medium	To Do	Unr
KBPS-19 FR-002: Cohort Registration	Unassigned	Shrihari Nagesh	Medium	To Do	Unr
KBPS-19 Story 1.1: Cohort Registration	Unassigned	Shrihari Nagesh	Medium	To Do	Unr
KBPS-16 Epic 2: Vaccination Scheduling & Administration	Unassigned	Shrihari Nagesh	Medium	To Do	Unr
KBPS-18 FR-001: Dose Interval Lock	Unassigned	Shrihari Nagesh	Medium	To Do	Unr
KBPS-10 FR-003: Center Slot Booking	Unassigned	Shrihari Nagesh	Medium	To Do	Unr
KBPS-11 FR-004: Dose Administration Check-in	Unassigned	Shrihari Nagesh	Medium	To Do	Unr
KBPS-20 Story 2.1: Dose Interval Lock	Unassigned	Shrihari Nagesh	Medium	To Do	Unr
KBPS-21 Story 2.3: Dose Administration Check-in	Unassigned	Shrihari Nagesh	Medium	To Do	Unr
KBPS-25 Story 2.2: Center Slot Booking	Unassigned	Shrihari Nagesh	Medium	To Do	Unr
KBPS-17 Epic 3: Certificate Generation & Data Security	Unassigned	Shrihari Nagesh	Medium	To Do	Unr
KBPS-12 FR-005: Digital Certificate Generation	Unassigned	Shrihari Nagesh	Medium	To Do	Unr
KBPS-13 NFR-001: Fast Certificate Verification	Unassigned	Shrihari Nagesh	Medium	To Do	Unr
KBPS-14 NFR-002: Secure Citizen Data	Unassigned	Shrihari Nagesh	Medium	To Do	Unr
KBPS-22 Story 3.1: Digital Certificate Generation	Unassigned	Shrihari Nagesh	Medium	To Do	Unr
KBPS-23 Story 3.2: Fast Certificate Verification	Unassigned	Shrihari Nagesh	Medium	To Do	Unr
KBPS-24 Story 3.3: Secure Citizen Data	Unassigned	Shrihari Nagesh	Medium	To Do	Unr

Figure 4. Story 1.1: Cohort Registration (KBPS-19) expanded, showing its 4 sub-tasks (KBPS-26 to KBPS-29): build the registration form UI, implement the cohort-eligibility rule engine, validate age/comorbidity inputs, and unit test cohort-assignment logic.

4. Backlog — Epics Assigned

Epic	Theme	Backlog Items (Stories)
Epic 1	Citizen Registration & Eligibility	Story 1.1 (FR-002)
Epic 2	Vaccination Scheduling & Administration	Story 2.1 (FR-001), Story 2.2 (FR-003), Story 2.3 (FR-004)

Epic	Theme	Backlog Items (Stories)
Epic 3	Certificate Generation & Data Security	Story 3.1 (FR-005), Story 3.2 (NFR-001), Story 3.3 (NFR-002)

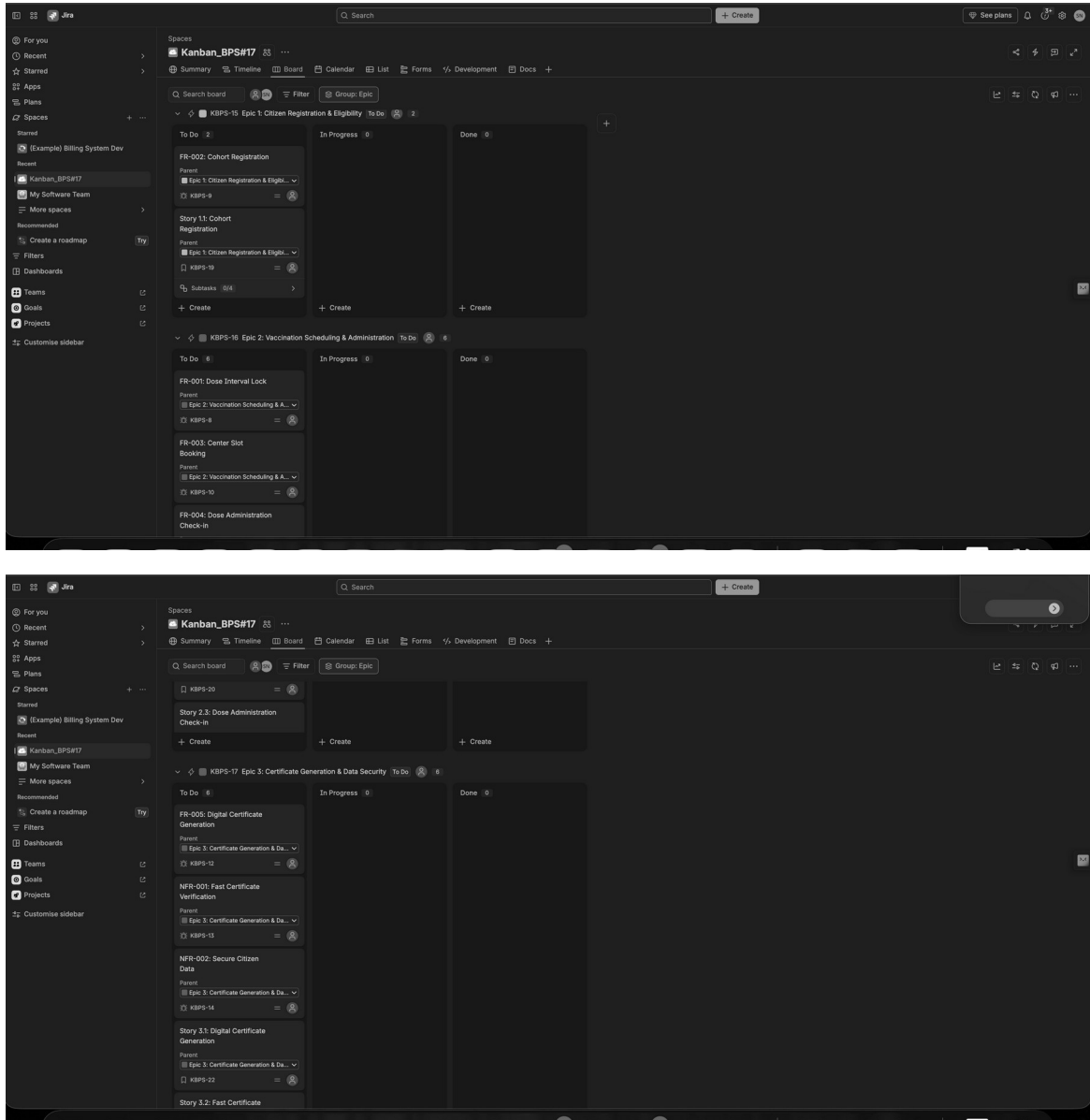


Figure 5. Full backlog (Board, Group: Epic) across two scrolled views — every work item assigned to its Epic: Epic 1 (FR-002, Story 1.1), Epic 2 (FR-001, FR-003, FR-004, Story 2.1, Story 2.3, and more below), Epic 3 (FR-005, NFR-001, NFR-002, Story 3.1, Story 3.2 and more below).